

SUBJECT: CSA0212 C PROGRAMMING

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OPERATORS AND EXPRESSIONS :

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, a 'Secure Exit' button, the lab name 'SIMATS Online Programming Lab', the subject 'C PROGRAMMING', and a user profile for 'NANDITHA S' with ID '192412467RB'. The left sidebar shows 'QUESTIONS 8 / 8 completed' and a dropdown menu with 'Swap 2 No's' selected. Below this is a section titled 'SWAP 2 NO'S' with a description: 'C Program to Swap Two Numbers. Swapping two numbers in C programming means swapping the values of two variables. For example, there are two variables m & n. Value of m is "2" & value of n is "3". Before Swapping: m value = 2; n value = 3. After Swapping: m value = 3; n value = 2'. The main area is the 'PROGRAM EDITOR' with a dropdown set to 'C' and buttons for 'Run' and 'Save'. It contains a C program for swapping two numbers. The 'OUTPUT' section shows the results of the program execution.

```
1 #include<stdio.h>
2 int main()
3 {
4     int a,b;
5     scanf("%d%d",&a,&b);
6     printf("Bfore swapping a=%db=%d\n",a,b);
7     a=a+b;
8     b=a-b;
9     a=a-b;
10    printf("After swapping a=%d b=%d",a,b);
11    return 0;
12 }
```

OUTPUT

```
Bfore swapping a=10b=20
After swapping a=-40 b=-30
```

INPUT

10 20

The screenshot shows the SIMATS Online Programming Lab interface. The header is identical to the previous one. The left sidebar shows 'QUESTIONS 8 / 8 completed' and a dropdown menu with 'Arithmetic' selected. Below this is a section titled 'ARITHMATIC' with a description: 'Write a program that takes two numbers as input and prints their sum, difference, product, and quotient.' The main area is the 'PROGRAM EDITOR' with a dropdown set to 'C' and buttons for 'Run' and 'Save'. It contains a C program for performing arithmetic operations. The 'OUTPUT' section shows the results of the program execution.

```
1 #include<stdio.h>
2 int main()
3 {
4     int a,b;
5     scanf("%d%d",&a,&b);
6     printf("add=",a+b);
7     printf("sub=",a-b);
8     printf("product=",a*b);
9     return 0;
10 }
```

OUTPUT

```
add=sub=product=
```

INPUT

2 4



Secure Exit

SIMATS Online Programming Lab

C PROGRAMMING

 NANDITHA S
192412467RB

QUESTIONS 8 / 8 completed

Convert

CONVERT

Write a program that converts Celsius to Fahrenheit. The formula is Fahrenheit = (Celsius * 1.8) + 32.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main ()
3 {
4     float celsius , fahrenheit;
5     printf("enter the temperature in celsius :");
6     scanf("%f",&celsius);
7     fahrenheit =(celsius*1.8)+32;
8     printf("tempreature in frahenheit=%.2f",fahrenheit);
9     return 0;
10 }
```

OUTPUT

enter the temperature in celsius :tempreature in frahenheit=208.40

INPUT

98



Secure Exit

SIMATS Online Programming Lab

C PROGRAMMING

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QUESTIONS 8 / 8 completed

Rectangle

RECTANGLE

Write a program that calculates the area of a rectangle given its length and width. Problem Description: Write a program that calculates the area of a rectangle given its length and width. The program should prompt the user to enter the length and width of the rectangle as input, then calculate the area and print it to the console.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     float length, width, area;
5     printf("entr the length:");
6     scanf("%f",&length);
7     printf("enter the width");
8     scanf("%f",&width);
9     area=length*width;
10    printf("areaof the rectangle=%2f",area);
11    return 0;
12 }
13 }
```

OUTPUT

entr the length:enter the width:areaof the rectangle=12.000000

INPUT

3 4 5



Secure Exit

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C PROGRAMMING

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QUESTIONS 8 / 8 completed

ODD EVEN

ODD EVEN

Write a program that checks whether a number is even or odd.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a;
5     printf("enter the values");
6     scanf("%d",&a);
7     if (a%2==0)
8         printf("a is even");
9     else
10        printf("a is odd");
11    return 0;
12 }
```

OUTPUT

enter the valuesa is even

INPUT

4



Secure Exit

SIMATS Online Programming Lab

C PROGRAMMING

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QUESTIONS 8 / 8 completed

Leap

LEAP

Write a program that determines whether a year is a leap year or not. A year is a leap year if it is divisible by 4, but not divisible by 100 unless it is also divisible by 400.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a ;
5     printf("enter the value:\n");
6     scanf("%d",&a);
7     if((a%4==0 && a%100!=0)|| a%100==0)
8         printf("a is leap year");
9     else
10        printf("a is not leap year");
11
12 }
```

OUTPUT

enter the value:
a is leap year

INPUT

2024



Secure Exit

SIMATS Online Programming Lab

C PROGRAMMING

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QUESTIONS 8 / 8 completed

Max-Min

MAX-MIN

Write a program that takes three integers as input and prints the maximum and minimum of the three.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a,b,c;
5     scanf("%d%d%d",&a,&b,&c);
6     if (a>b && a>c)
7         printf("a is the greatest");
8     else if (b>c)
9         printf("b is the greatest");
10    else
11        printf("c is the greatest");
12    printf("\n");
13    if (a<b && a<c)
14        printf("a is the smallest");
15    else if (b<c)
16        printf("b is smallest");
17    else
18        printf("c is smallest");
19    return 0;
20 }
21 }
```

OUTPUT

c is the greatest
a is the smallest

INPUT

1 2 3



Secure Exit

SIMATS Online Programming Lab

C PROGRAMMING

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QUESTIONS 8 / 8 completed

Sum-Avg

SUM-AVG

Write a program that takes a list of numbers as input and computes the sum and average of the numbers.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a,b,c,sum=0;
5     float avg;
6     scanf("%d%d%d",&a,&b,&c);
7     sum=a+b+c;
8     avg=sum/3.0;
9     printf("sum%d avg%2f",sum,avg);
10    return 0;
11 }
12 }
```

OUTPUT

sum1681922473 avg560640832.000000

INPUT

30 40 50